

Unsupervised Learning and Evol. Computation

Using R

Summer Term 2026

Case Study

The Case Study consists of three exercises. For each exercise, write a short report on your analysis and findings and submit a respective .pdf-file. Be as precise as possible but make sure to include only the most relevant findings. Usage of the provided **Latex template is mandatory**. Also, hand in your R-Code as a **meaningfully commented .R-file**.

Please, work in groups of three persons. You have to **assign yourself to a group** in PANDA until **June 14th, 23:59**. Otherwise, you are unable to submit your case study. In your reports, you are supposed to add a statement of contribution to indicate which person did which parts of the case study.

Make sure that your submitted files are recognisably different from the solutions of your colleagues! You are only allowed to use LLMs to improve your writing style!

Your submissions will not receive a regular numerical grade. Instead, they will be assessed using four categories: FAILED, PASSED, GOOD, and EXCELLENT. You must achieve at least PASSED to be allowed to take the exam. A submission will be assessed as PASSED if it earns at least 50% (30) of the total points. If your submission is assessed as GOOD or EXCELLENT, you can receive a bonus for the final exam grade:

GOOD your final grade improves by one grade step.

EXCELLENT your final grade improves by two grade steps.

Please note that this bonus only applies if you pass the exam. Exercises marked with (★) have a strong influence on the bonus category. For these exercises, please focus on meaningful findings rather than simply providing lengthy answers.

Please, upload your results until **July 5th, 23:59** in PANDA. It is sufficient that only one person of your group uploads the results.

1 Volleyball Match Statistics

[20]

The dataset *matchStats.csv* contains detailed statistics from **international volleyball matches** played in **July and August 2025**. Each row represents a single match between two national teams, including performance metrics (kills, blocks, aces, digs, etc.) and the **set-by-set results**.

A **volleyball match** is played between **two teams**, each aiming to score points by grounding the ball on the opponent's side of the court under organized rules. Here's how it typically works:

- A match is played in a **best-of-five sets** format.
- To **win a set**, a team must reach **25 points** (with at least a 2-point lead).
- If both teams win two sets each, a **fifth "tiebreak" set** (to **15 points**) is played.
- Teams score points in several ways:
 - **Kills**: When a team attacks successfully, landing the ball in the opponent's court.
 - **Blocks**: When a defensive play stops or deflects an opponent's attack.
 - **Aces**: When a served ball directly results in a point.
 - **Opponent errors**: When the other team makes a fault or mistake.
- The team that wins **three sets first** is the **match winner**.

Column	Description
Date	Date of the match (YYYY-MM-DD).
Home Team	The team designated as playing at home.
Away Team	The team designated as visiting.
Winner	The team that won the match.
Loser	The team that lost the match.
Kills Home / Away	Successful attack points made by each team.
Blocks Home / Away	Points scored via successful blocks.
Aces Home / Away	Points from unreturned serves.
Opponents Errors Home / Away	Points gained from the opponent's mistakes.
Total Points Home / Away	Total points scored by each team in all sets combined.
Digs Home / Away	Defensive actions preventing the ball from touching the court.
Receptions Home / Away	Successful receptions of serves.
Set1–Set5 Home / Away	Points scored by each team in each set. If fewer than five sets were played, remaining values are blank.

Exercises

1. Load the dataset *matchStats.csv*.
2. Check for columns with missing values. For each column, state which data missing mechanism is likely present and give a reason for your decision.
3. For each column, decide on a suited imputation technique and argue why you chose this technique. Create a dataset with all values filled.
4. Add a column indicating whether the home team won (1 = home team won; 0 = home team lost).
5. Pick a column and check if it is normally distributed. Use QQ plot and a suited statistical test for normality.
6. (★) Conduct some exploratory data analysis on your own, e.g.:
 - relationships between variables,
 - empirical distribution and anomalies in the data,
 - differences between home and away teams or winner and loser teams.

You do not have to analyze all of this comprehensively. Pick out interesting aspects. Generate suitable visualizations with `ggplot2` to support your analysis.

2 Volleyball Players Statistics

[20]

The dataset *playerStats.csv* contains **player-level performance metrics** from professional international volleyball matches. Each row represents the match statistics of an **individual player**, summarizing their contributions across various phases of play – setting, receiving, serving, blocking, defending, and attacking.

In volleyball, each position has specialized responsibilities:

- **Setter** – orchestrates offensive plays by delivering accurate sets to attackers.
- **Outside Hitter (Left Wing Spiker)** – primary attacker who occasionally receives serves and plays defense.
- **Opposite Spiker** – attacking from the right side.
- **Middle Blocker** – specializes in blocking and quick attacks near the net.
- **Libero** – defensive specialist focused on receptions and digs.

Column	Description
Player Name	Full name or abbreviated name of the player.
Team	Country or club the player belongs.
Position	Player's on-court role (e.g., OPPOSITE SPIKER, OUTSIDE HITTER).
Age	Player's age in years.
Height	Player's height (in centimeters).
Setting Errors	Errors committed during setting attempts.
Sets Per Match	Average number of sets per match.
Successful Receives	Number of successfully received serves.
Receiving Errors	Unsuccessful or faulty receptions.
Receives Per Match	Average receptions per match.
Aces	Serves that directly result in a point.
Service Errors	Faults committed during serving.
Service Attempts	Total number of serves attempted.
Serves Per Match	Average number of serves per match.
Blocks	Successful blocks resulting in points.
Blocking Errors	Unsuccessful or faulted block attempts.
Blocks Per Match	Average blocks per match.
Defensive Errors	Mistakes made during defensive plays.
Defensive Receptions	Total number of opponent attacks received.
Digs Per Match	Average successful digs per match.
Kills	Attacks that directly score a point.
Attacking Errors	Errors made during attack attempts.
Attacking Attempts	Total number of attack (spike) attempts.
Attacks Per Match	Average number of attacks per match.

Exercises

1. Load the dataset `playerStats.csv`.
2. Create a column that sums up all errors for each player (they all end with `Errors`).
3. Change height to numeric, e.g., from 203cm to 203.
4. (★) Apply exploratory data analysis. Get an overview of the number and kind of variables, compute descriptive statistics, visualize your data, and check for correlations.
5. Check for uni- and multivariate outliers.
6. Group the data per country and calculate the mean of all numeric columns. Compare these mean values with the team statistics found in `teamStats.csv`.
7. For each position, calculate the best player in terms of total errors and the worst player in terms of total errors. Store the best and worst players in a single tibble, and add a column indicating whether the player is the best or worst for their position.

3 Volleyball Players Clustering

[20]

For this exercise, use the dataset from the previous exercise.

Exercises

1. Load the dataset `playerStats.csv`. Again, change height to numeric, e.g., from `203cm` to `203`.
2. (★) Try to find homogeneous groups of athletes by clustering on suitable variables. Do **not** use name, team, and position as clustering criteria, and do **not** use DBSCAN here. Decide on a suited number of clusters and provide an argument for your choice. Use an appropriate way to visualize your cluster results.
3. Compare your cluster results with the DBSCAN algorithm. Do the results differ? Again, use an appropriate way to visualize the results.
4. Analyze how your clusters align with the positions and countries.